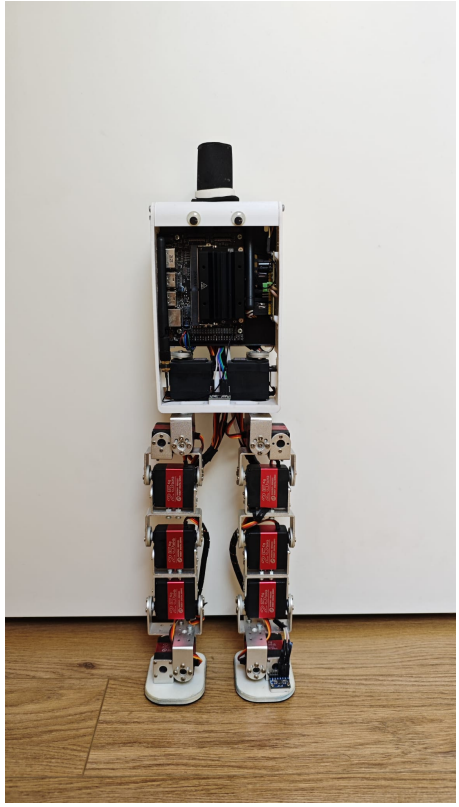
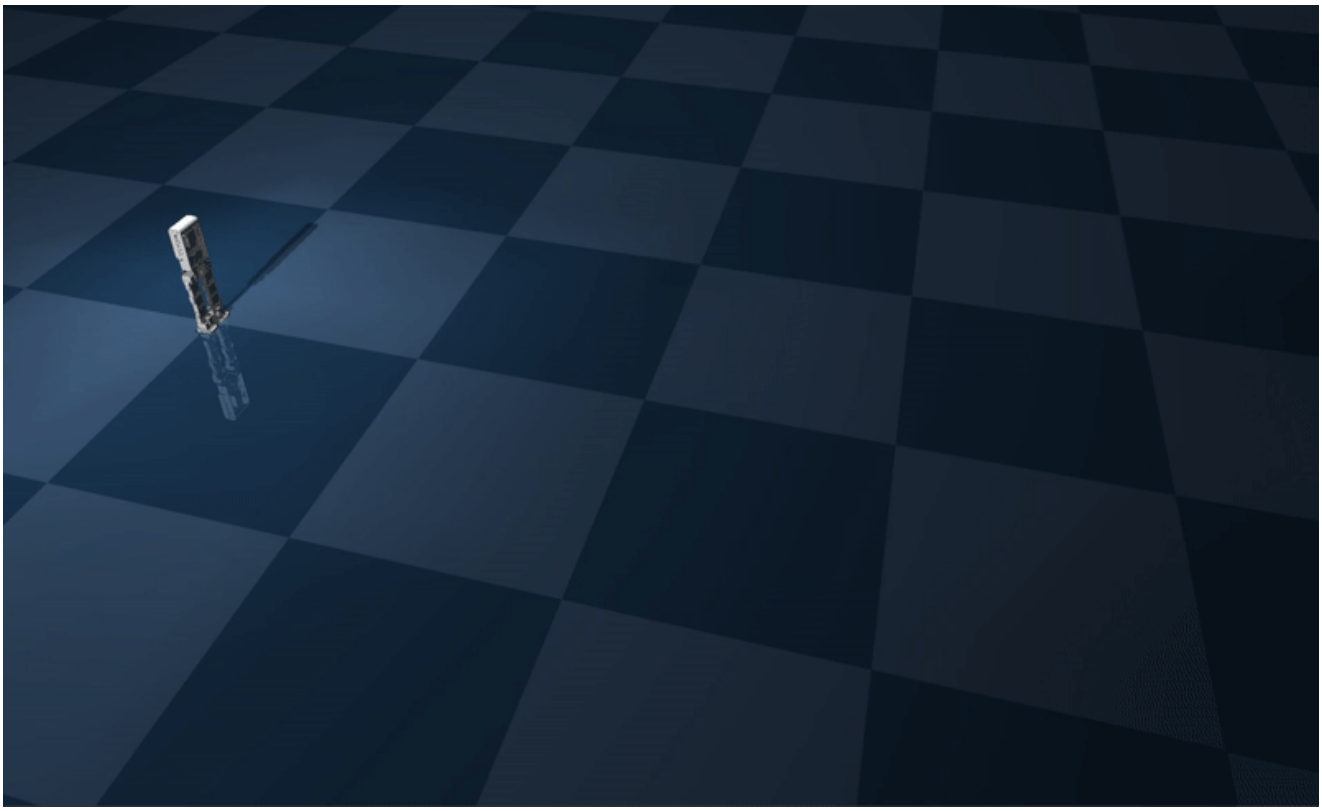
BipedRobot





Left: the physical robot. Right: learned gait in simulation.

About

Started as a hand-animated biped running on Arduino - servos driven by hardcoded angle sequences, no feedback, no learning. Over several years it evolved into a full reinforcement learning system built around a custom-made physical robot.

The primary environment is Isaac Lab (4096 parallel envs, fully on GPU end-to-end - simulation, inference, and training all in torch). MuJoCo is also supported and shares the same observation space, action space, and reward functions, so policies transfer between the two.

Everything was written from scratch: the RL algorithms, the neural network backbones, the environments, the reward functions, and the motion imitation pipeline.

RL algorithms	PPO, SAC, FastSAC, DDPG, D4PG, MPO
Architectures	MLP, CNN, LSTM, Causal Transformer - swappable via config
Simulation	Isaac Lab (primary) + MuJoCo (secondary, identical interface)
Observations	48-dim proprioceptive, asymmetric actor-critic with privileged critic obs
Reward	18 configurable terms shared between both environments
Motion imitation	FBX → NPZ pipeline, imitation reward terms for reference tracking
Sim-to-real	Physics randomization (mass, friction, damping) + left-right obs mirroring

Quick Start

```
# Clone with submodules
git clone --recurse-submodules <repo-url>
cd BipedRobot

# Install dependencies
pip install -r requirements.txt

# Isaac Lab training (primary)
python src/isaaclab/train.py --task BipedRobot \
    --config_path config/isaac/config_working.yaml

# MuJoCo training (no Isaac Lab required)
python src/mujoco/train.py --config config/final/train_config_sac.yaml
```

For Isaac Lab installation, see Isaac Lab setup below.

Project Structure

```
BipedRobot/
│
```

```
├── config/                        # Training configuration (YAML +
Pydantic schema)
│   ├── schema.py                  # All config fields with types and
defaults
│   ├── final/                     # Production configs (sac, ddpg,
d4pg, mpo)
│   └── isaac/                     # Isaac Lab-specific configs
│
├── envs/
│   ├── mujoco/                    # MuJoCo environment
│   │   ├── mujoco_env.py          # MujocoEnv - main Gymnasium
environment
│   │   ├── base_env.py            # GLFW rendering, contact helpers
│   │   └── distributed.py         # Multi-worker distribution
(Sequential / Parallel)
│   │
│   ├── isaaclab/                  # Isaac Lab GPU environment
│   │   ├── biped_env.py           # BipedEnv (DirectRLEnv, 4096
parallel envs)
│   │   ├── biped_env_cfg.py       # BipedEnvCfg, BipedRobotV2EnvCfg
│   │   ├── rewards/rewards.py     # Shared reward functions (pure
PyTorch)
│   │   └── mdp/                   # Commands and terrain logic
│   │
│   ├── rewards/
│   │   └── mujoco_reward.py       # MuJoCo reward - calls shared
rewards.py
│   │
│   ├── utils/
│   │   ├── mirroring.py           # Left-right obs/action mirroring
│   │   └── randomizer.py          # Physics randomization
│   │
│   └── assets/                    # Robot model files
│       ├── robot/                 # V1 robot (MuJoCo XML + URDF + USD)
│       └── robotV2/               # V2 robot
│
├── src/
│   ├── mujoco/                    # MuJoCo training entry points
│   │   ├── train.py               # Train (pass --checkpoint to resume)
│   │   ├── train_lstm.py          # Train with LSTM backbone
│   │   └── test_model.py          # Evaluate a saved model
│   │
│   ├── isaaclab/                  # Isaac Lab training entry points
│   │   ├── train.py               # Train (pass --checkpoint to resume)
│   │   ├── play.py                # Evaluate / play
│   │   └── common.py              # Shared IsaacLabWrapper base class
│   │
│   └── tools/                     # Developer utilities
│       ├── plot_symmetry_logs.py  # Plot symmetry error CSVs
│       ├── try_symmetry.py        # Symmetry validator (standalone)
│       └── isaaclab/              # Isaac Lab dev tools
│           ├── debug_env.py       # Interactive joint controller
(Tkinter)
```

```
│           ├── test_ik.py         # Interactive IK testing (Tkinter)
│           └── try_symmetry.py    # Symmetry validator with TensorBoard
│
├── utils/
│   ├── __init__.py                # Config, NoConfig classes
│   ├── ik_utils.py                # Multi-end-effector IK (Isaac Lab)
│   ├── motion_reference.py        # Motion reference loader (imitation
learning)
│   └── free_camera_movement.py    # MuJoCo free camera
│
├── torch-rl-algorithms/           # Submodule: RL algorithms + neural
backbones
├── robot-motion-reference/        # Submodule: FBX → NPZ motion
pipeline
│
└── requirements.txt
```

> See `envs/README.md` for environment details, how to add robots, and how to add reward terms.
> See `config/README.md` for the full configuration reference.
> See `src/README.md` for every runnable script with all arguments.

Setup

Dependencies

```
pip install -r requirements.txt
```

`requirements.txt` includes `-e torch-rl-algorithms/` - the submodule is installed in editable mode, so changes take effect immediately without reinstalling.

Submodules

Submodule	What it is
`torch-rl-algorithms`	PPO, SAC, DDPG, D4PG, MPO algorithms with MLP/CNN/LSTM/Transformer backbones, written from scratch
`robot-motion-reference`	Converts FBX motion capture files to NPZ joint trajectory arrays for use as imitation learning targets

Isaac Lab Setup

Install following the official guide. Isaac Lab provides `isaaclab`, `isaaclab_tasks`, and the PhysX simulator.

Training

Isaac Lab

```
python src/isaaclab/train.py --task BipedRobot \
    --config_path config/isaac/config_working.yaml
```

Argument	Default	Description
`--task`	`BipedRobot`	`BipedRobot` (V1) or `BipedRobotV2` (V2)
`--config_path`	`config/train_config.yaml`	Path to YAML config
`--num_envs`	from config	Override number of parallel environments
`--checkpoint`	None	Resume from checkpoint path
`--headless`	False	Run without GUI
`--video`	False	Record video during training

```
# Resume
python src/isaaclab/train.py --task BipedRobot \
    --checkpoint checkpoints/my_run/step_500000.pt

# Play / evaluate
python src/isaaclab/play.py --task BipedRobot \
    --config_path config/isaac/config_working.yaml \
    --checkpoint checkpoints/my_run/step_1000000.pt
```

MuJoCo

```
python src/mujoco/train.py --config config/final/train_config_sac.yaml
```

Argument	Default	Description
`--config`	`config/final/train_config_d4pg.yaml`	Path to YAML config
`--checkpoint`	None	Trainer state directory to resume from

```
# Resume
python src/mujoco/train.py --config config/final/train_config_sac.yaml \
    --checkpoint checkpoints/sac/my_run/trainer_state/

# Evaluate
python src/mujoco/test_model.py --config
config/final/train_config_sac.yaml \
```

```
    --checkpoint checkpoints_final/sac/step_30000000.pt

# Train with LSTM backbone
python src/mujoco/train_lstm.py --config
config/final/train_config_mpo.yaml
```

Environments

Both environments expose the same interface: 48-dim observations, normalized `[-1, 1]^12` actions, identical reward functions. Policies transfer between them.

Property	Isaac Lab	MuJoCo
Parallelism	4096 envs	Sequential or multiprocess
Obs (actor)	48 dims	48 dims (identical)
Obs (critic)	actor + contact scalar	actor + forces + friction + damping + masses
Action space	`Box[-1, 1]^12`	`Box[-1, 1]^12`
Reward	`envs/isaaclab/rewards/rewards.py`	`envs/rewards/mujoco_reward.py`
Reward functions	**Shared** - pure PyTorch, called from both envs	MuJoCo converts numpy→tensor
Termination	Tilt > 45° or both feet airborne	Tilt > 45° or height < 0.15 m

Observation layout (48 dims):

```
[lin_acc (3)] [ang_vel_body (3)] [projected_gravity (3)] [commands (3)]
[joint_pos_rel (12)] [joint_vel (12)] [prev_actions (12)]
```

For full environment documentation - how to add robots, modify observations, add reward terms - see `envs/README.md`.

Configuration

Configs are YAML files validated against the Pydantic schema in `config/schema.py`. Algorithm, architecture, environment, reward weights, physics randomization - all in one file.

Minimal example:

```yaml
train:
  model: "sac"              # sac | ddpg | d4pg | mpo | ppo
  steps: 3.0e7
```

```
    worker_groups: 8
    workers_per_group: 16
    seed: 42
    log_dir: "runs"
    checkpoint_path: "checkpoints/"
    model_name: "my_run"
    actor_obs: "normal"
    critic_obs: "privileged"

  model:
    actor_config:
      network_type: "mlp"    # mlp | cnn | lstm | transformer
      hidden_sizes: [256, 256]
    critic_config:
      network_type: "mlp"
      hidden_sizes: [256, 256]
    actor_lr: 1.0e-4
    critic_lr: 1.0e-4
```

See `config/final/` for production-tested algorithm configs, `config/isaac/` for Isaac Lab configs, and `config/README.md` for the full field reference.

Motion Imitation

The `robot-motion-reference` submodule provides a pipeline to convert FBX motion capture animations into NPZ joint trajectory files. These can be used in Isaac Lab training as imitation reward targets via `track_joint_pos_exp` and `track_joint_vel_exp` reward terms.

```
# In Isaac Lab config:
env_config:
  animation_npz_path: "robot-motion-reference/output/walk.npz"
  animation_loop: true
  animation_speed: 1.0
  animation_random_start: true
rewards:
  track_joint_pos_exp: 1.0
  track_joint_vel_exp: 0.5
```

See the robot-motion-reference submodule for the FBX conversion pipeline.

Developer Tools

Script	Requires Isaac Lab	Description

Script	Requires Isaac Lab	Description
`src/tools/isaaclab/debug_env.py`	Yes	Tkinter UI - drive joints with sliders, inspect root height and contact forces
`src/tools/isaaclab/test_ik.py`	Yes	Tkinter UI - set foot IK targets with 6 sliders per foot
`src/tools/isaaclab/try_symmetry.py`	Yes	Symmetry validator - logs left/right divergence to TensorBoard
`src/tools/try_symmetry.py`	No	Standalone symmetry validator (no Isaac Lab required)
`src/tools/plot_symmetry_logs.py`	No	Plot symmetry error CSVs from symmetry tests

```
# Interactive joint controller
python src/tools/isaaclab/debug_env.py --task BipedRobot

# Interactive IK test
python src/tools/isaaclab/test_ik.py --task BipedRobot

# Symmetry validation (no Isaac Lab)
python src/tools/try_symmetry.py --checkpoint
checkpoints/my_run/step_1000000.pt
```

Checkpoints

Saved to `checkpoints/<model_name>/step_<N>.pt` during training. Full trainer state (replay buffer + optimizers) is saved to `checkpoints/<model_name>/trainer_state/` for exact resume.

```
checkpoints/
└── my_run/
    ├── step_100000.pt        ← weights only, for evaluation/transfer
    ├── step_200000.pt
    └── trainer_state/        ← full state for resume (replay buffer,
optimizers, step count)
```

License

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